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Elektronika Analog

LATIHAN SOAL BAB 5

1. a. What is the expected amplification of a BJT transistor amplifier if the dc supply is set to zero volts?

b. What will happen to the output ac signal if the dc level is insufficient? Sketch the effect on the waveform.

c. What is the conversion efficiency of an amplifier in which the effective value of the current through a 2.2-k load is 5 mA and the drain on the 18-V dc supply is 3.8 mA?

(a) If the dc power supply is set to zero volts, the amplification will be zero.

(b) Too low a dc level will result in a clipped output waveform.

(c) $P_o = I^2R = (5\text{ mA})^2 2.2\text{ k}\Omega = 55\text{ mW}$

$P_i = V_{CC}I = (18\text{ V})(3.8\text{ mA}) = 68.4\text{ mW}$

$$\eta = \frac{P_o(\text{ac})}{P_i(\text{dc})} = \frac{55\text{ mW}}{68.4\text{ mW}} = 0.804 \Rightarrow \mathbf{80.4\%}$$

2. Can you think of an analogy that would explain the importance of the dc level on the resulting ac gain?

3. If a transistor amplifier has more than one dc source, can the superposition theorem be applied to obtain the response of each dc source and algebraically add the results?

No, because they intersect to define the Q point.

4. What is the reactance of a 10μF capacitor at a frequency of 1 kHz? For networks in which the resistor levels are typically in the kilohm range, is it a good assumption to use the short-circuit equivalence for the conditions just described? How about at 100 kHz?

$$X_C = \frac{1}{2\pi fC} = \frac{1}{2\pi(1\text{ kHz})(10\ \mu\text{F})} = \mathbf{15.92\ \Omega}$$

$f = 100$ kHz: $X_C = \mathbf{0.159\ \Omega}$

Yes, better at 100 kHz

5. Given the common-base configuration of Fig. 5.150 , sketch the ac equivalent using the notation for the transistor model appearing in Fig. 5.7 .

5.4 The r_e Transistor Model

6. a. Given an Early voltage of $V_A = 100$ V, determine r_o if $V_{CE_Q} = 8$ V and $I_{C_Q} = 4$ mA.

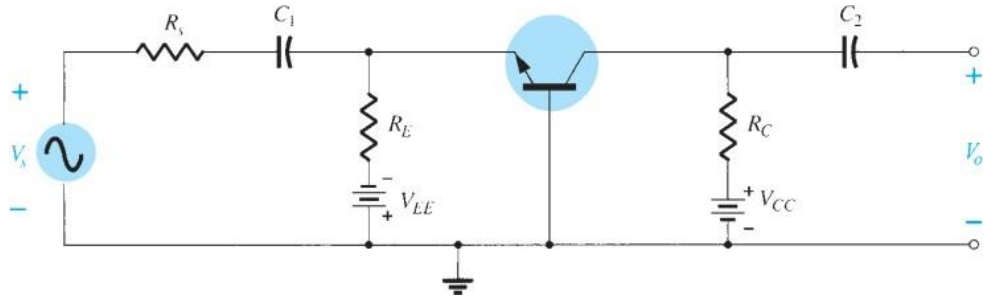
b. Using the results of part (a), find the change in I_C for a change in V_{CE} of 6 V at the same Q-point as part (a).

(a) $$r_o = \frac{V_A + V_{CE_Q}}{I_{C_Q}} = \frac{100\text{ V} + 8\text{ V}}{4\text{ mA}} = \mathbf{27\ k\Omega}$$

(b) $$r_o = \frac{\Delta V_{CE}}{\Delta I_C} \Rightarrow \Delta I_C = \frac{\Delta V_{CE}}{r_o} = \frac{6\text{ V}}{r_o} = \frac{6\text{ V}}{27\text{ k}\Omega} = \mathbf{0.222\ mA}$$

7. For the common-base configuration of Fig. 5.18, an ac signal of 10 mV is applied, resulting in an ac emitter current of 0.5 mA. If $\alpha = 0.980$, determine:

a. Z_i.
b. V_o if $R_L = 1.2\ \text{k}\Omega$.
c. $A_v = V_o/V_i$.
d. Z_o with $r_o = \infty\ \Omega$.
e. $A_i = I_o/I_i$.
f. I_b.



10. For the common-base configuration of Fig. 5. l8, the dc emitter current is 3.2 mA and a is 0.99. Determine the following if the applied voltage is 48 mV and the load is 2.2 kn.

a. rz.

b. Z,. if

e. *A* .

d. A,, if ft, = 1.2 kf1.

$$y = \frac{48}{\quad}$$

(a) r — ' "" -- i‹n

$I;$ 3.2 mA

(b) $Z, — r, — 15$ f2

(c) $I_z — cd, —$ (0.99)(3.2 mA) = 3.168 mA

(d) Vy - !c *i* - (3.168 mA)(2.2 kn) - 6•97 V

(e) $A — \frac{V}{V,} = \frac{6.97\ V}{48\ mV} = 145.21$

(f) $Iz —$ (1 — @(— (1 — 0.99)i. = (0.01)(3.2 mA)

=S2

$$A_i = \frac{\overline{r_o + R_L} \cdot \beta I_b}{I_b} = \frac{r_o}{r_o + R_L} \cdot \beta$$

$$\frac{40\ k\Omega}{40\ k\Omega + 1.2\ k\Omega} (30)$$

= 77.67

d A - ∘ , r, = 1.2 kE2)40 k0

"" " r, " " t3 El

1.165 kf2

13 f2

9. The input impedance to a common-emitter transistor amplifier is 1. 2 kf1 with Q 14t l,

$rp —$ so kn, md «, - 2.7 kf1. Determine:

b. *If* if *V;* — 3O m V.

d. $A_i = I_o/I_i = I_L/I_b$.
e. $A_v = V_o/V_i$.

(a) $2i$ —- $Qr‹$ —- (l40)r, = 1200

$$gg_{-} \frac{1200}{} - 8.57 \text{ f2}$$

(b) $I - \frac{V,}{Z;} \frac{30 \text{ mV}}{1.2 \text{ kf2}} = 25,\text{«A}$

(c) $I,$. -- *fill* -- (140)(25 ,uA) = 3.5 mA

(d) $I_L = \frac{r\,I}{r + Rz} \frac{(50 \text{ kf2})(3.5 \text{ mA})}{50 \text{ kf2} + 2.7 \text{ kf2}} = 3.321\text{mA}$

$A, — \frac{3.321 \text{ mA}}{25\,\mu\text{A}} = 132.84$

(e) $A, — \frac{V,}{V_i} = \frac{Z}{1.2 \text{ k£2}} = —(132.84)(2.7 \text{ kf2})$

$= —298.59$

11. For the network of Fig. 5.151 :

a. Determine Zi and Zo .

b. Find Av .

c. Repeat parts (a) and (b) with ro = 20 kΩ.

Jawab :

$$A_v = -\frac{R_C}{r_e} \Rightarrow r_e = -\frac{R_C}{A_v} = -\frac{4.7\ \text{k}\Omega}{(-200)} = 23.5\ \Omega$$

$$r_e = \frac{26\ \text{mV}}{I_E} \Rightarrow I_E = \frac{26\ \text{mV}}{r_e} = \frac{26\ \text{mV}}{23.5\ \Omega} = 1.106\ \text{mA}$$

$$I_B = \frac{I_E}{\beta + 1} = \frac{1.106\ \text{mA}}{91} = 12.15\ \mu\text{A}$$

$$I_B = \frac{V_{CC} - V_{BE}}{R_B} \Rightarrow V_{CC} = I_B R_B + V_{BE}$$

$$= (12.15\ \mu\text{A})(1\ \text{M}\Omega) + 0.7\ \text{V}$$

$$= 12.15\ \text{V} + 0.7\ \text{V}$$

$$= \mathbf{12.85\ V}$$

12. For the network of Fig. 5.152 , determine VCC for a voltage gain of Av = -160.

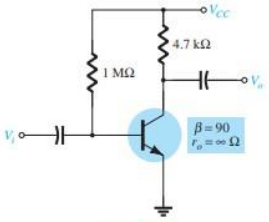


FIG. 5.152
Problem 12.

(a) r_e: $I_B = \dfrac{V_{CC} - V_{BE}}{R_B} = \dfrac{12\ \text{V} - }{22($

$I_E = (\beta + 1)I_B = (60 + 1)(51.36\ \mu\text{A})$

$= 3.13\ \text{mA}$

$$r_e = \frac{26\ \text{mV}}{I_E} = \frac{26\ \text{mV}}{3.13\ \text{mA}} = 8.31\ \Omega$$

$Z_i = R_B \| \beta r_e = 220\ \text{k}\Omega \| (60)(8.31\ \Omega) = 220\ \text{k}\Omega \| 498.6\ \Omega$

$= \mathbf{497.47\ \Omega}$

$r_o \geq 10R_C \therefore Z_o = R_C = \mathbf{2.2\ k\Omega}$

(b) $$A_v = -\frac{R_C}{r_e} = \frac{-2.2\ \text{k}\Omega}{8.31\ \Omega} = \mathbf{-264.74}$$

(c) $Z_i = \mathbf{497.47\ \Omega}$ (the same)

$Z_o = r_o \| R_C = 20\ \text{k}\Omega \| 2.2\ \text{k}\Omega$

$= \mathbf{1.98\ k\Omega}$

$$A_v = \frac{-R_C \| r_o}{r_e} = \frac{-1.98\ \text{k}\Omega}{8.31\ \Omega} = \mathbf{-238.27}$$

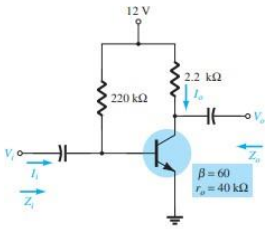


FIG. 5.151
Problem 11.

Jawab :

13. For the network of Fig. 5.153 :

a. Calculate IB, IC , and re .
b. Determine Zi and Zo .
c. Calculate Av .
d. Determine the effect of ro = 30 kΩ on Av .

14. For r the network of Fig. 5.153 , what value of RC will cut the voltage gain to half the value obtaine

Jawab :

15. For the network of Fig. 5.154
    a. Determine r_e .
    b. Calculate Z_i and Z_o .
    c. Find A_v .
    d. Repeat parts (b) and (c) with ro = 25 kΩ.

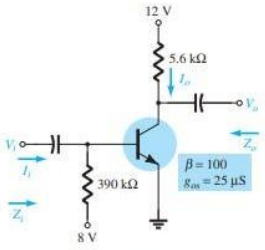


FIG. 5.153
Problem 13.

(a) $I_B = \dfrac{V_{CC} - V_{BE}}{R_B} = \dfrac{8\text{ V} - 0.7\text{ V}}{390\text{ k}\Omega} = \mathbf{18.72\ \mu A}$

$I_E = (\beta + 1)I_B = (101)(18.72\ \mu\text{A}) = 1.89\text{ mA}$

$r_e = \dfrac{26\text{ mV}}{I_E} = \dfrac{26\text{ mV}}{1.89\text{ mA}} = \mathbf{13.76\ \Omega}$

$I_C = \beta I_B = (100)(18.72\ \mu\text{A}) = \mathbf{1.87\ mA}$

(b) $Z_i = R_B \| \beta r_e = 390\text{ k}\Omega \| (100)(13.76\ \Omega) = 390\text{ k}\Omega \| 1.38\text{ k}\Omega$
$= \mathbf{1.38\ k\Omega}$

$r_o \geq 10R_C \therefore Z_o = R_C = \mathbf{5.6\ k\Omega}$

(c) $A_v = -\dfrac{R_C}{r_e} = \dfrac{-5.6\text{ k}\Omega}{13.76\ \Omega} = \mathbf{-406.98}$

(d) $A_v = -\dfrac{R_C \| r_o}{r_e} = -\dfrac{(5.6\text{ k}\Omega) \| (30\text{ k}\Omega)}{13.76\ \Omega} = -\dfrac{4.72\text{ k}\Omega}{13.76\ \Omega} = \mathbf{-343.03}$

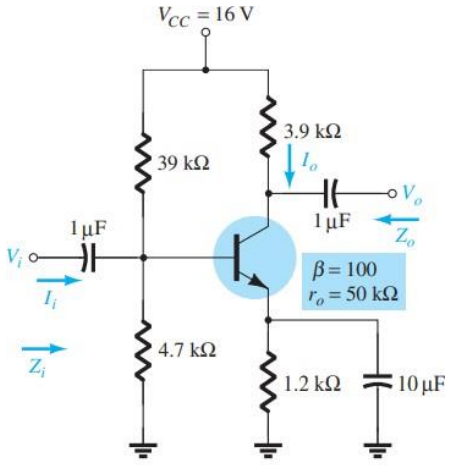


FIG. 5.154

Jawab:

a.

$$(100)(1.2\ \text{k}\Omega) \geq 10(4.7\ \text{k}\Omega)$$
$$120\ \text{k}\Omega > 47\ \text{k}\Omega \text{ (satisfied)}$$

Use approximate approach:

$$V_B = \frac{R_2 V_{CC}}{R_1 + R_2} = \frac{4.7\ \text{k}\Omega(16\ \text{V})}{39\ \text{k}\Omega + 4.7\ \text{k}\Omega} = 1.721\ \text{V}$$

$$V_E = V_B - V_{BE} = 1.721\ \text{V} - 0.7\ \text{V} = 1.021\ \text{V}$$

$$I_E = \frac{V_E}{R_E} = \frac{1.021\ \text{V}}{1.2\ \text{k}\Omega} = 0.8507\ \text{mA}$$

$$r_e = \frac{26\ \text{mV}}{I_E} = \frac{26\ \text{mV}}{0.8507\ \text{mA}} = \mathbf{30.56\ \Omega}$$

b.

$$Z_i = R_1 \parallel R_2 \parallel \beta r_e$$
$$= 4.7\ \text{k}\Omega \parallel 39\ \text{k}\Omega \parallel (100)(30.56\ \Omega)$$
$$= \mathbf{1.77\ k\Omega}$$
$$r_o \geq 10R_C \therefore Z_o \cong R_C = \mathbf{3.9\ k\Omega}$$

c.

$$A_v = -\frac{R_C}{r_e} = -\frac{3.9\ \text{k}\Omega}{30.56\ \Omega} = \mathbf{-127.6}$$

d.

$r_o = 25\ \text{k}\Omega$

(b) $Z_i(\text{unchanged}) = \mathbf{1.77\ k\Omega}$

$Z_o = R_C \parallel r_o = 3.9\ \text{k}\Omega \parallel 25\ \text{k}\Omega = \mathbf{3.37\ k\Omega}$

(c) $$A_v = -\frac{(R_C \parallel r_o)}{r_e} = -\frac{(3.9\ \text{k}\Omega)\parallel(25\ \text{k}\Omega)}{30.56\ \Omega} = -\frac{3.37\ \text{k}\Omega}{30.56\ \Omega}$$

$= \mathbf{-110.28}$ (vs. −127.6)

16. Determine V_{cc} for the network of Fig. 5.155 if A_v=-160 and r_o= 100 kΩ.

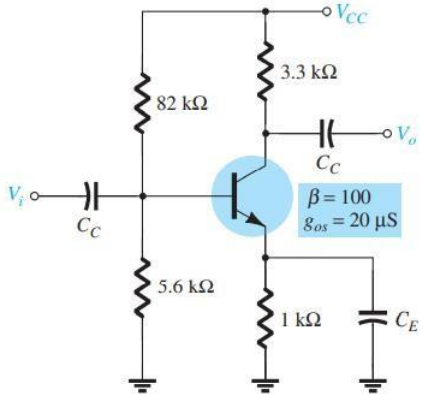


FIG. 5.155

Jawab:

$\beta R_E \geq 10R_2$

$(100)(1\ \text{k}\Omega) \geq 10(5.6\ \text{k}\Omega)$

$100\ \text{k}\Omega > 56\ \text{k}\Omega$ (checks!) & $r_o \geq 10R_C \Rightarrow 50\ \text{k}\Omega \geq 10(3.3\text{k}\Omega) = 33\ \text{k}\Omega$

Use approximate approach:

$$A_v = -\frac{R_C}{r_e} \Rightarrow r_e = -\frac{R_C}{A_v} = -\frac{3.3\ \text{k}\Omega}{-160} = \mathbf{20.625\ \Omega}$$

$$r_e = \frac{26\ \text{mV}}{I_E} \Rightarrow I_E = \frac{26\ \text{mV}}{r_e} = \frac{26\ \text{mV}}{20.625\ \Omega} = 1.261\ \text{mA}$$

$$I_E = \frac{V_E}{R_E} \Rightarrow V_E = I_E R_E = (1.261\ \text{mA})(1\ \text{k}\Omega) = 1.261\ \text{V}$$

$$V_B = V_{BE} + V_E = 0.7\ \text{V} + 1.261\ \text{V} = 1.961\ \text{V}$$

$$V_B = \frac{5.6\ \text{k}\Omega\, V_{CC}}{5.6\ \text{k}\Omega + 82\ \text{k}\Omega} = 1.961\ \text{V}$$

$$5.6\ \text{k}\Omega\, V_{CC} = (1.961\ \text{V})(87.6\ \text{k}\Omega)$$

$$V_{CC} = \mathbf{30.68\ V}$$

17. For the network of Fig. 5.156 :

a. Determine r_e.

b. Calculate V_b and V_c.

c. Determine Z_i and A_v = Vo>Vi.

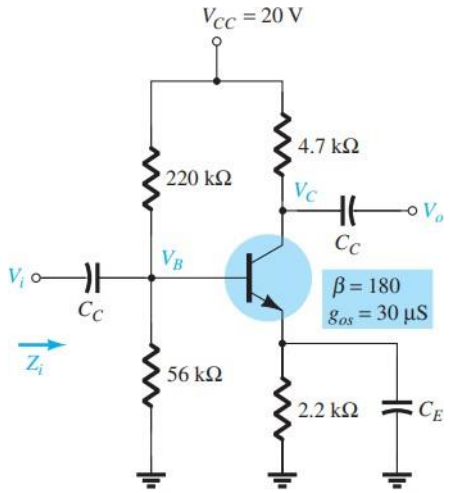


FIG. 5.156

Jawab:

a.

Test $\beta R_E \geq 10R_2$

$$(180)(2.2\ \text{k}\Omega) \overset{?}{\geq} 10(56\ \text{k}\Omega)$$

$396\ \text{k}\Omega < 560\ \text{k}\Omega$ (not satisfied)

Use exact analysis:

(a) $R_{Th} = 56\ \text{k}\Omega \parallel 220\ \text{k}\Omega = 44.64\ \text{k}\Omega$

$$E_{Th} = \frac{56\ \text{k}\Omega(20\ \text{V})}{220\ \text{k}\Omega + 56\ \text{k}\Omega} = 4.058\ \text{V}$$

$$I_B = \frac{E_{Th} - V_{BE}}{R_{Th} + (\beta+1)R_E} = \frac{4.058\ \text{V} - 0.7\ \text{V}}{44.64\ \text{k}\Omega + (181)(2.2\ \text{k}\Omega)}$$

$$= 7.58\ \mu\text{A}$$

$$I_E = (\beta + 1)I_B = (181)(7.58\ \mu\text{A})$$

$$= 1.372\ \text{mA}$$

$$r_e = \frac{26\ \text{mV}}{I_E} = \frac{26\ \text{mV}}{1.372\ \text{mA}} = \mathbf{18.95\ \Omega}$$

b.

$$V_E = I_E R_E = (1.372\ \text{mA})(2.2\ \text{k}\Omega) = 3.02\ \text{V}$$

$$V_B = V_E + V_{BE} = 3.02\ \text{V} + 0.7\ \text{V}$$

$$= \mathbf{3.72\ V}$$

$$V_C = V_{CC} - I_C R_C$$

$$= 20\ \text{V} - \beta I_B R_C = 20\ \text{V} - (180)(7.58\ \mu\text{A})(4.7\ \text{k}\Omega)$$

$$= \mathbf{13.59\ V}$$

C,

$$Z_i = R_1 \parallel R_2 \parallel \beta r_e$$

$$= 56\ \text{k}\Omega \parallel 220\ \text{k}\Omega \parallel (180)(18.95\ \text{k}\Omega)$$

$$= 44.64\ \text{k}\Omega \parallel 3.41\ \text{k}\Omega$$

$$= \mathbf{3.17\ k\Omega}$$

$$r_o < 10R_C$$

$33.33\ \text{k}\Omega < 10(4.7\ \text{k}\Omega) = 47\ \text{k}\Omega$ (checks)

$$\therefore A_v = -\frac{R_C \parallel r_o}{r_e}$$

$$= -\frac{(6.8\ \text{k}\Omega) \parallel (50\ \text{k}\Omega)}{18.95\ \Omega}$$

$$= \mathbf{-298.15}$$

18. For the network of Fig. 5.157 :
    a. Determine r_e.
    b. Find the dc voltages V_B, V_{CB}, and V_C.
    c. Determine Z_i and Z_o.
    d. Calculate A_v ——————————— V o>V i .

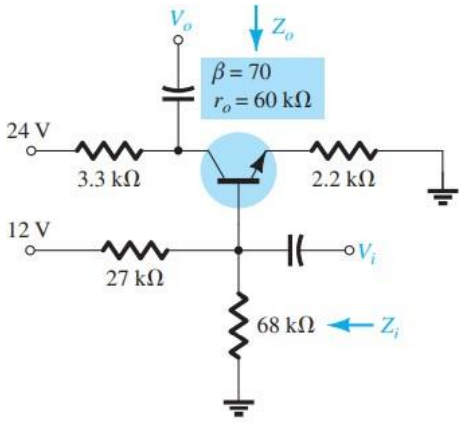


FIG. 5.157

Jawab:

a.

$$R_{Th} = 27\ \text{k}\Omega \,||\, 68\ \text{k}\Omega = 19.32\ \text{k}\Omega$$

$$E_{Th} = \frac{68\ \text{k}\Omega(12\ \text{V})}{68\ \text{k}\Omega + 27\ \text{k}\Omega} = 8.59\ \text{V}$$

$$I_B = \frac{E_{Th} - V_{BE}}{R_{Th} + (\beta+1)R_E} = \frac{9.59\ \text{V} - 0.7\ \text{V}}{19.32\ \text{k}\Omega + (70+1)(2.2\ \text{k}\Omega)}$$

$$= 44.95\ \mu\text{A}$$

$$I_E = (\beta+1)I_B = (70+1)(44.95\ \mu\text{A}) = 3.19\ \text{mA}$$

$$r_e = \frac{26\ \text{mV}}{I_E} = \frac{26\ \text{mV}}{3.19\ \text{mA}} = \mathbf{8.15\ \Omega}$$

b.

$$V_B = V_{BE} + I_E R_E = 0.7\ \text{V} + (3.19\ \text{mA})(2.2\ \text{k}\Omega)$$

$$= 7.72\ \text{V}$$

$$V_C = 24\ \text{V} - I_C R_C = 24\ \text{V} - (\beta I_B)(3.3\ \text{k}\Omega)$$
$$= 24\ \text{V} - (70)(44.95\ \mu\text{A})(3.3\ \text{k}\Omega)$$
$$= \mathbf{13.61\ V}$$

$$V_{CB} = V_C - V_B = 13.61\ \text{V} - 7.72\ \text{V} = \mathbf{5.89\ V}$$
$$V_E = V_B - 0.7 = 7.72\ \text{V} - 0.7\ \text{V} = \mathbf{7.02\ V}$$
$$V_{CE} = V_C - V_E = 13.61\ \text{V} - 7.02\ \text{V} = \mathbf{6.59\ V}$$

c.

$$Z_i = 68\ \text{k}\Omega \,||\, 27\ \text{k}\Omega \,||\, \beta R_E = 19.3\ \text{k}\Omega \,||\, (70)(2.2\ \text{k}\Omega)$$
$$= 19.32\ \text{k}\Omega \,||\, 154\ \text{k}\Omega = \mathbf{17.17\ k\Omega}$$

d.

$$A_v = \frac{-R_C}{R_E} = -\frac{3.3\ \text{k}\Omega}{2.2\ \text{k}\Omega} = \mathbf{-1.5}$$

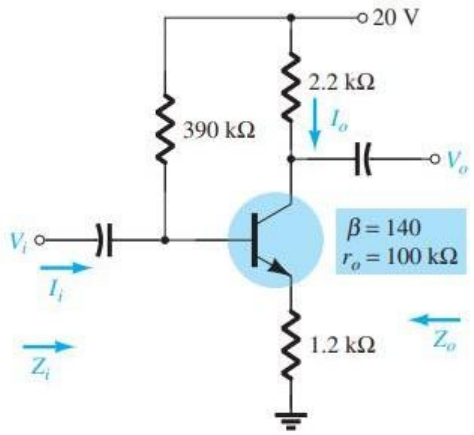


FIG. 5.158
Problems 19 and 20.

19. For the network of Fig. 5.158 :
    a. Determine r e .
    b. Find Z i and Z o .
    c. Calculate A v .
    d. Repeat parts (b) and (c) with ro = 20 k ohm.

(a) $I_B = \dfrac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_E} = \dfrac{20\text{ V} - 0.7\text{ V}}{390\text{ k}\Omega + (141)(1.2\text{ k}\Omega)}$

$= \dfrac{19.3\text{ V}}{559.2\text{ k}\Omega} = 34.51\ \mu\text{A}$

$I_E = (\beta + 1)I_B = (140 + 1)(34.51\ \mu\text{A}) = 4.866\text{ mA}$

$r_e = \dfrac{26\text{ mV}}{I_E} = \dfrac{26\text{ mV}}{4.866\text{ mA}} = \mathbf{5.34\ \Omega}$

(b) $Z_b = \beta r_e + (\beta + 1)R_E$

$= (140)(5.34\text{ k}\Omega) + (140 + 1)(1.2\text{ k}\Omega) = 747.6\ \Omega + 169.9\text{ k}\Omega$

$= \mathbf{169.95\ k\Omega}$

$Z_i = R_B \parallel Z_b = 390\text{ k}\Omega \parallel 169.95\text{ k}\Omega = \mathbf{118.37\ k\Omega}$

$Z_o = R_C = \mathbf{2.2\ k\Omega}$

(c) $A_v = -\dfrac{\beta R_C}{Z_b} = -\dfrac{(140)(2.2\text{ k}\Omega)}{169.95\text{ k}\Omega} = \mathbf{-1.81}$

(d) $Z_b = \beta r_e + \left[\dfrac{(\beta + 1) + R_C / r_o}{1 + (R_C + R_E)/r_o}\right] R_E$

$= 747.6\ \Omega \left[\dfrac{(141) + 2.2\text{ k}\Omega / 20\text{ k}\Omega}{1 + (3.4\text{ k}\Omega)/20\text{ k}\Omega}\right] 1.2\text{ k}\Omega$

$= 747.6\ \Omega + 144.72\text{ k}\Omega$

$= 145.47\text{ k}\Omega$

$Z_i = R_B \parallel Z_b = 390\text{ k}\Omega \parallel 145.47\text{ k}\Omega = \mathbf{105.95\ k\Omega}$

$Z_o = R_C = \mathbf{2.2\ k\Omega}$ (any level of r_o)

$$A_v = \frac{V_o}{V_i} = \frac{-\dfrac{\beta R_C}{Z_b}\left[1 + \dfrac{r_e}{r_o}\right] + \dfrac{R_C}{r_o}}{1 + \dfrac{R_C}{r_o}}$$

$$= \frac{\dfrac{-(140)(2.2\text{ k}\Omega)}{145.47\text{ k}\Omega}\left[1 + \dfrac{5.34\ \Omega}{20\text{ k}\Omega}\right] + \dfrac{2.2\text{ k}\Omega}{20\text{ k}\Omega}}{1 + \dfrac{2.2\text{ k}\Omega}{20\text{ k}\Omega}}$$

$$= \frac{-2.117 + 0.11}{1.11} = \mathbf{-1.81}$$

20. Repeat Problem 19 with R E bypassed. Compare results.

(a) dc analysis the same

$\therefore\ r_e = \mathbf{5.34\ \Omega}$ (as in #19)

(b) $Z_i = R_B \parallel Z_b = R_B \parallel \beta r_e = 390\text{ k}\Omega \parallel (140)(5.34\ \Omega) = \mathbf{746.17}\ \Omega$ vs. 118.37 kΩ in #32

$Z_o = R_C = \mathbf{2.2\ k\Omega}$ (as in #32)

(c) $A_v = \dfrac{-R_C}{r_e} = \dfrac{-2.2\text{ k}\Omega}{5.34\ \Omega} = \mathbf{-411.99}$ **vs** −1.81 in #19

(d) $Z_i = \mathbf{746.17}\ \Omega$ vs. 105.95 kΩ for #19

$Z_o = R_C \parallel r_o = 2.2\text{ k}\Omega \parallel 20\text{ k}\Omega = \mathbf{1.98\ k\Omega}$ vs. 2.2 kΩ in #19

$A_v = -\dfrac{R_C \parallel r_o}{r_e} = -\dfrac{1.98\text{ k}\Omega}{5.34\ \Omega} = \mathbf{-370.79}$ vs. −1.81 in #19

Significant difference in the results for A_v.

21. For the network of Fig. 5.159 , determine R E and R B if Av = -10 and re = 3.8 ohm. Assume that Zb = bRE.

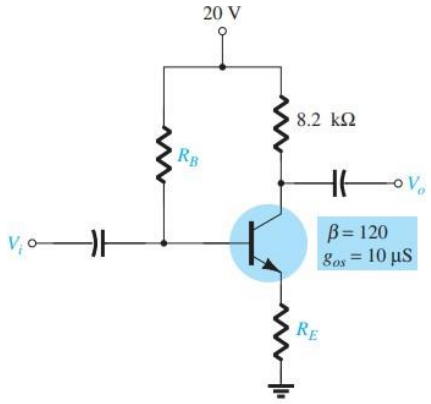


FIG. 5.159
Problem 21.

$r_o \geq 10R_C$
$100\ \text{k}\Omega \geq 10(8.2\ \text{k}\Omega) = 82\ \text{k}\Omega$
Therefore:

$$A_v = -\frac{\beta R_C}{Z_b} = -\frac{\beta R_C}{\beta R_E} = -\frac{R_C}{R_E} = -10$$

$$\therefore R_E = \frac{R_C}{10} = \frac{8.2\ \text{k}\Omega}{10} = \mathbf{0.82\ k\Omega}$$

$$I_E = \frac{26\ \text{mV}}{r_e} = \frac{26\ \text{mV}}{3.8\ \Omega} = 6.842\ \text{mA}$$

$V_E = I_E R_E = (6.842\ \text{mA})(0.82\ \text{k}\Omega) = 5.61\ \text{V}$
$V_B = V_E + V_{BE} = 5.61\ \text{V} + 0.7\ \text{V} = 6.31\ \text{V}$

$$I_B = \frac{I_E}{(\beta+1)} = \frac{6.842\ \text{mA}}{121} = 56.55\ \mu\text{A}$$

$$\text{and } R_B = \frac{V_{R_B}}{I_B} = \frac{V_{CC} - V_B}{I_B} = \frac{20\ \text{V} - 6.31\ \text{V}}{56.55\ \mu\text{A}} = \mathbf{242.09\ k\Omega}$$

22. For the network of Fig. 5.160 :
    a. Determine r e .
    b. Find Z i and A v .

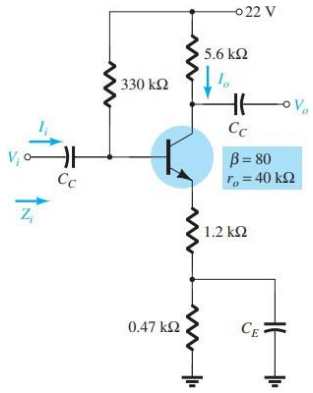


FIG. 5.160
Problem 22.

□

(a) $$I_B = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_E}$$

$$= \frac{22\text{ V} - 0.7\text{ V}}{330\text{ k}\Omega + (81)(1.2\text{ k}\Omega + 0.47\text{ k}\Omega)} = \frac{21.3\text{ V}}{465.27\text{ k}\Omega}$$

$$= 45.78\ \mu\text{A}$$

$$I_E = (\beta + 1)I_B = (81)(45.78\ \mu\text{A}) = 3.71\text{ mA}$$

$$r_e = \frac{26\text{ mV}}{I_E} = \frac{26\text{ mV}}{3.71\text{ mA}} = \mathbf{7\ \Omega}$$

(b) $r_o < 10(R_C + R_E)$

$$\therefore Z_b = \beta r_e + \left[\frac{(\beta + 1) + R_C / r_o}{1 + (R_C + R_E)/r_o}\right] R_E$$

$$= (80)(7\ \Omega) + \left[\frac{(81) + 5.6\text{ k}\Omega / 40\text{ k}\Omega}{1 + 6.8\text{ k}\Omega / 40\text{ k}\Omega}\right] 1.2\text{ k}\Omega$$

$$= 560\ \Omega + \left[\frac{81 + 0.14}{1 + 0.17}\right] 1.2\text{ k}\Omega$$

(note that $(\beta + 1) = 81 \gg R_C/r_o = 0.14$)

$$= 560\ \Omega + [81.14\ /1.17]1.2\text{ k}\Omega = 560\ \Omega + 83.22\text{ k}\Omega$$

$$= \mathbf{83.78\ k\Omega}$$

$$Z_i = R_B \parallel Z_b = 330\text{ k}\Omega \parallel 83.78\text{ k}\Omega = \mathbf{66.82\ k\Omega}$$

$$A_v = \frac{\dfrac{-\beta R_C}{Z_b}\left(1 + \dfrac{r_e}{r_o}\right) + \dfrac{R_C}{r_o}}{1 + \dfrac{R_C}{r_o}}$$

$$= \frac{\dfrac{-(80)(5.6\text{ k}\Omega)}{83.78\text{ k}\Omega}\left(1 + \dfrac{7\ \Omega}{40\text{ k}\Omega}\right) + \dfrac{5.6\text{ k}\Omega}{40\text{ k}\Omega}}{1 + 5.6\text{ k}\Omega/40\text{ k}\Omega}$$

$$= \frac{-(5.35) + 0.14}{1 + 0.14}$$

$$= \mathbf{-4.57}$$

$$7, = \text{"} \frac{\quad}{\quad} = \frac{\quad \text{”}}{2} = \text{—}203.49$$

“ r

@ = (—203.49j{l3.76 LI) = 26 kO